

# EE256, Power and Energy

(2025\_version)

Fundamentals of Alternating Current (AC) systems:

Importance of AC:

When High voltage DC (HVDC) generation is not possible, ac provides a better means for transmitting power over large distances with lower losses. Because the generated voltages, which is typically at 20 kV, can be easily raised to higher voltages such as 132 kV, 220 kV (as in Sri Lanka) for transmission purposes simply by using transformers. In contrast, raising DC to Higher DC voltages (HVDC) require power electronics. However, for larger distances HVDC transmission is having a cost advantage and becoming popular for many reasons now.

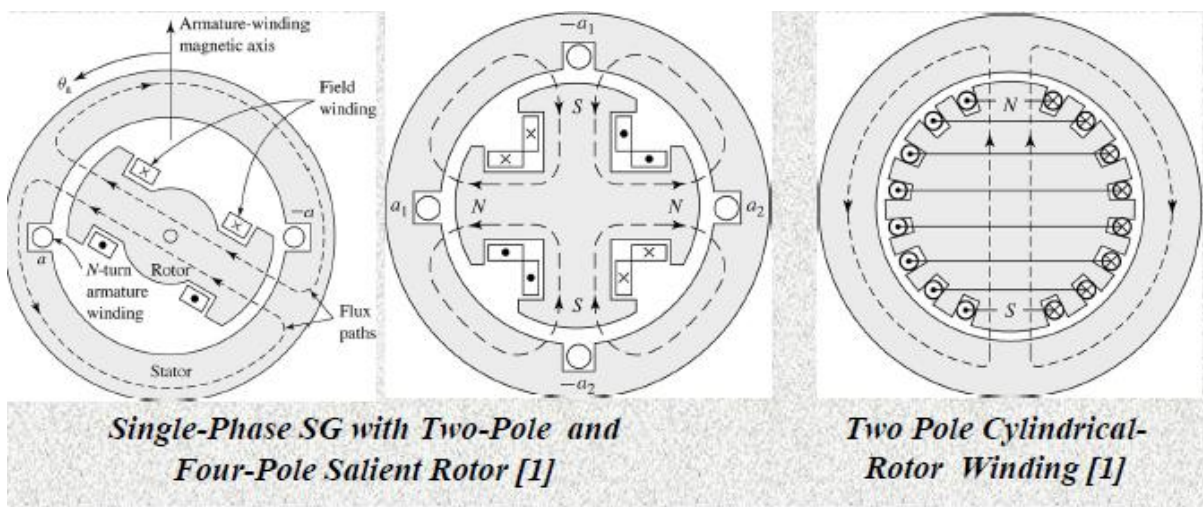
History:

"In 1888 alternating current systems gained further viability with introduction of a functional AC motor, something these systems had lacked up till then. The design, an induction motor, was independently invented by **Galileo Ferraris** and **Nikola Tesla** (with Tesla's design being licensed by Westinghouse in the US)"- *extracted from World Wide Web.*

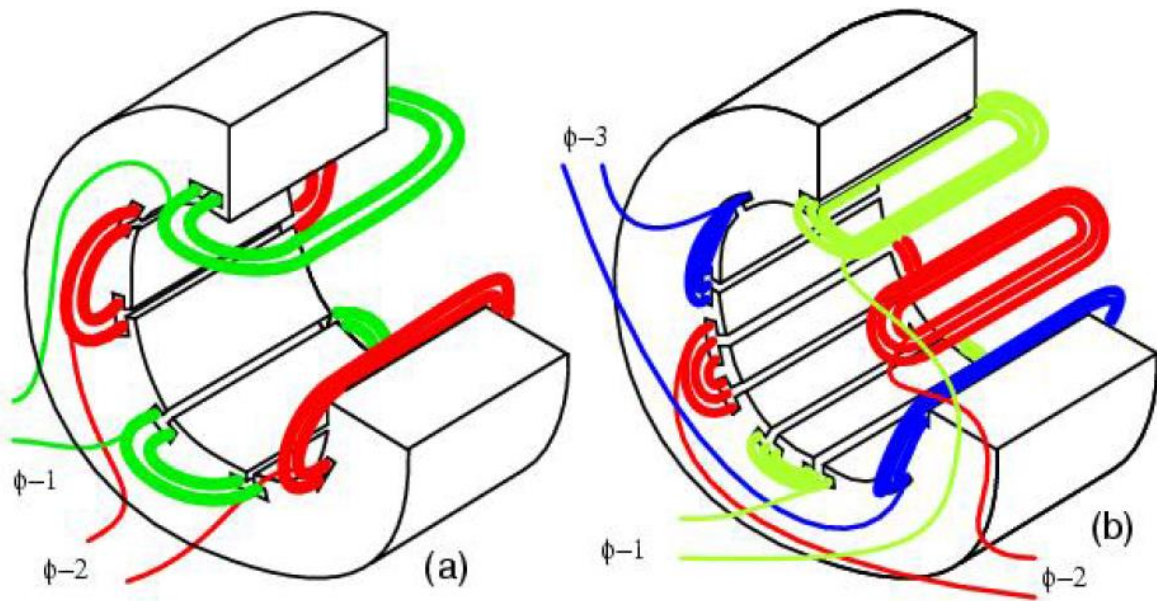
Nikola Tesla, invented the three-phase system too. 3-phase system has many exciting features which we go into details soon.



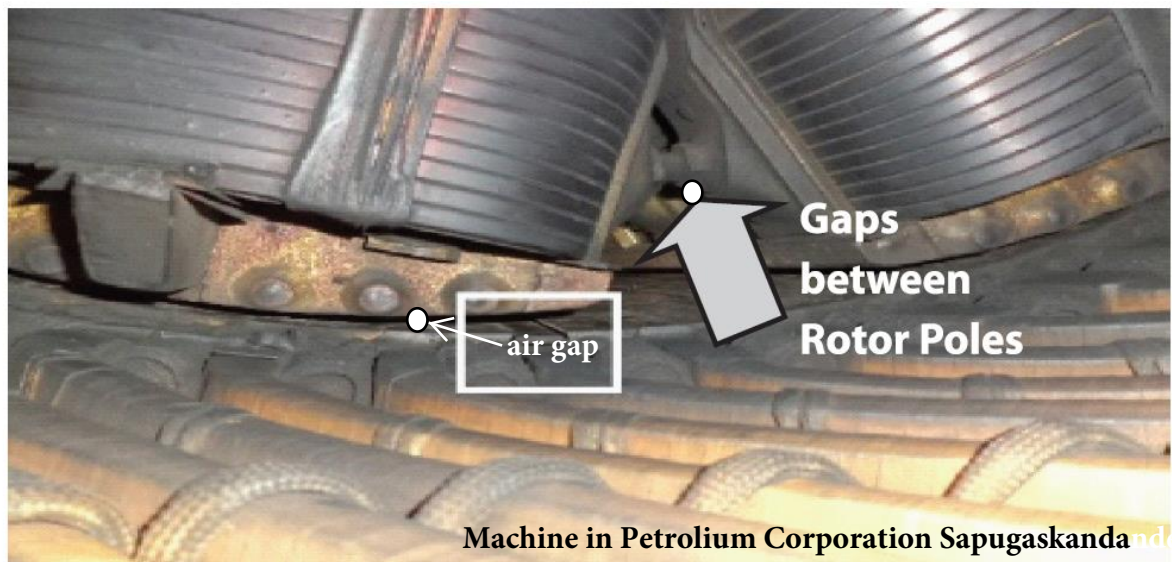
Synchronous generators: Winding Arrangements:



### Example of Two Phase and 3-phase Windings



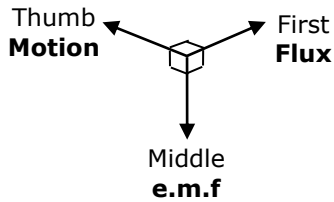
### Synchronous Generator: Stator and Rotor



## REVIEW OF BASIC MAGNETICS

### 1. Flemings Right-hand Rule

To figure out the direction of induced e.m.f. on a moving conductor in a magnetic field.



### 2. Lenz's Law:

*"The direction of an induced e.m.f. is always such that it tends to set up a current opposing the change of flux responsible for inducing that e.m.f."*

### 3. Right-hand Rule

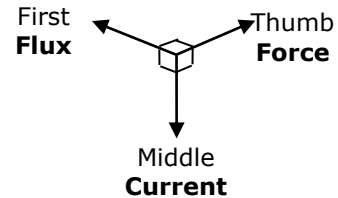
To figure out the direction of magnetic field around a current carrying conductor

### 4. Lorentz's Force Equation

$$\vec{F} = q(\vec{E} + (\vec{v} \times \vec{B}))$$

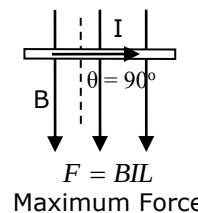
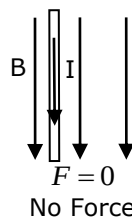
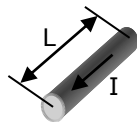
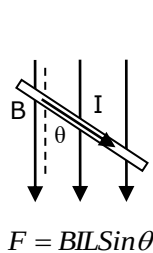
### 5. Flemings Left-hand Rule

To figure out the direction of force exerted on a current carrying conductor in a magnetic field



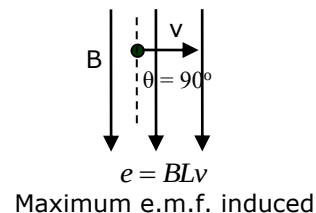
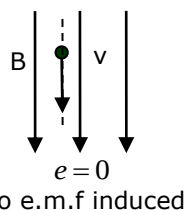
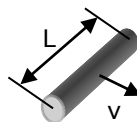
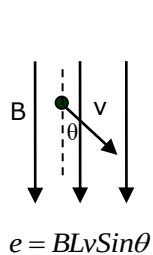
## Force on a current carrying conductor in a Magnetic field

For electrical machines, only magnetic field is effective. Force  $\vec{F} = q(0 + (\vec{v} \times \vec{B}))$ . Consider a conductor of length L carrying a current I A.



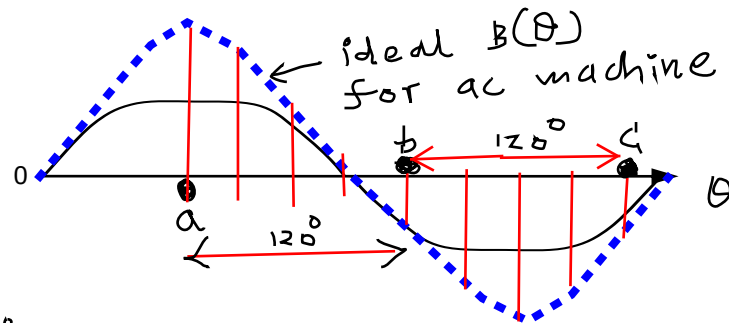
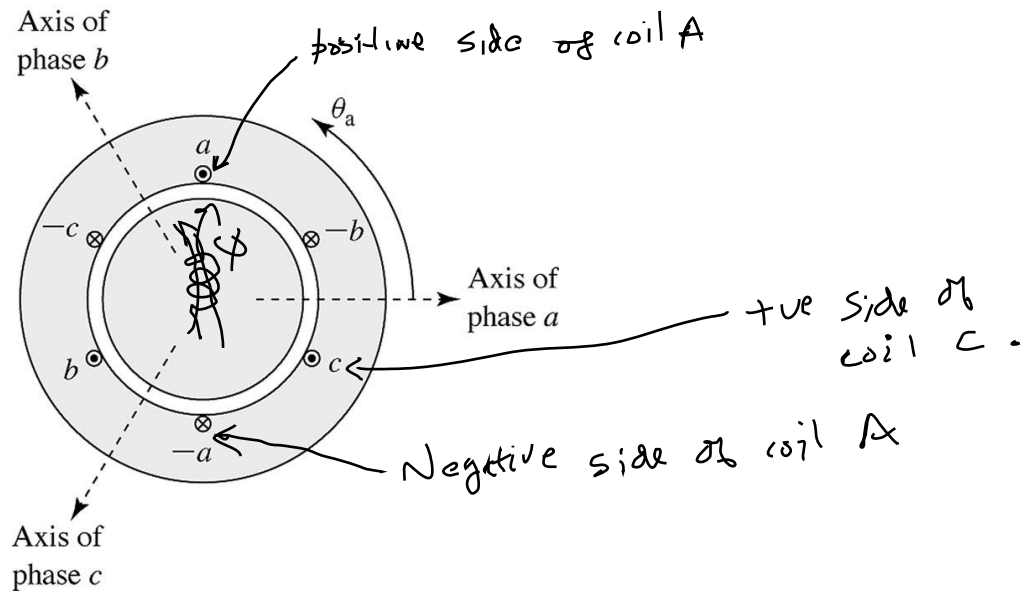
- Fleming's LH rule gives the direction of force
- Typically for PM,  $B = 0.1\text{T}$ . If  $L = 1\text{m}$  and  $I = 1\text{A}$ ,  
 $F = \text{????}$

## Electromotive force (e.m.f.), e, in a moving conductor in a Magnetic field.



- Fleming's RH rule gives the direction of induced e.m.f.
- $e = \text{Flux cut by the conductor per second}$ .
- Electrical Motors and Generators use the principles explained above. The conductors (windings) of them are designed to get the highest force (for motors) and e.m.f. (for generators) possible out of the system.

Since the conductors for phase A,B and C are placed 120 degrees (Electrical) apart to each other, the emf generated are 120 electrical degrees apart. The figure below illustrates the physical placement of concentrated coils for each phase. The arrows normal to the plane of each coil represent the axis of each coil. Note that the axes are 120 degrees apart to each other and so are the relevant conductors on each coil.



Note:

$$e \propto B(\theta)$$

## 3-Phase System

We are familiar with 1-phase system and the **phasor** representation. In summary, The 3-phase system can be considered as, three-single-phase systems having 120 degree phase separation to each other, combined together. We studied how the 3-phases can be formed in practice. However, 3-phase system has many added advantages than having 3-sources together. In terms of the equipment sizing and quality of power, 3-phase system is superior to single phase.

**3-Phase system is inveted by Nikola Tesla (1856-1943), who invented the popular induction motor.**

Fig. L2.1, shows a 3-phase currents flowing in a 3-phase load which is supplied by a balanced three phase source. Note that the currents are 120 degrees apart and so should be the source voltages.

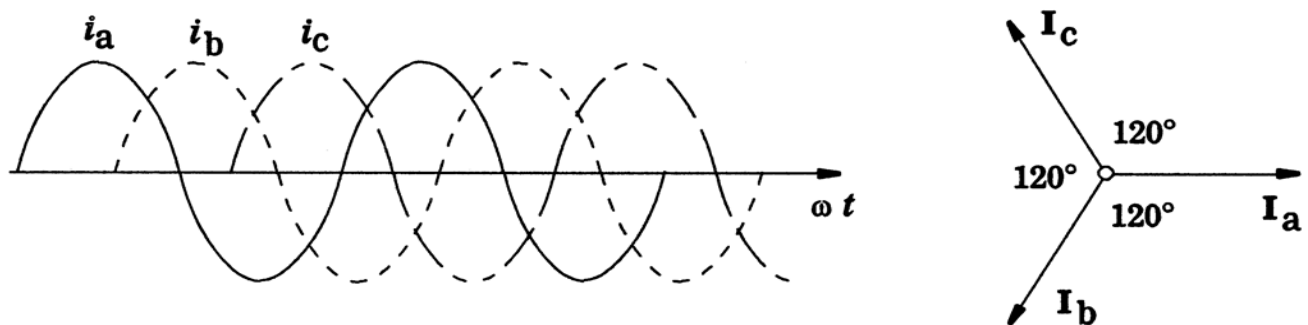
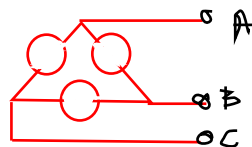
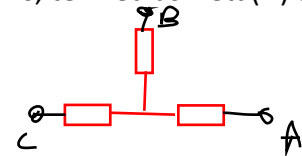


Fig. L2.1

3-phase sources/loads can be connected in two different configurations, termed as Delta( $\Delta$ ) and Star (Y).



Delta connected source



Star connected Load

Fig. L2.2 below shows a star-connected source, feeding a delta-connected load. Any combination of load/source selected from delta or star is possible. However, customers are connected using a transformer with star connected secondary (Y connected **source**). A Load may be star or delta connected. Usually combination of single phase loads and 3-phase loads are common thus making the system unbalance. However, utilities take care to balance the load on 3-phases. This way neutral current can be made to be almost zero. We will see some examples later.

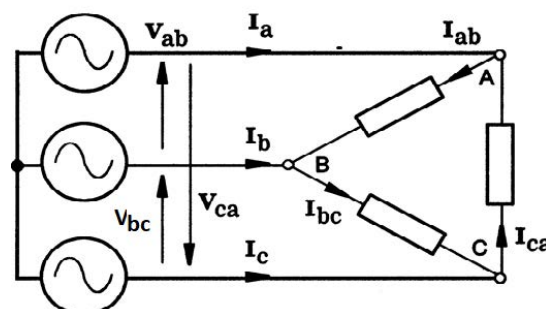


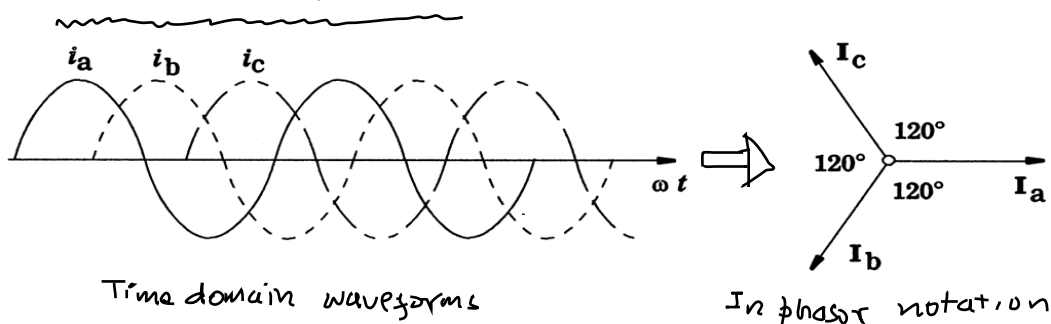
Fig. L2.2

## Phasors and Phase Sequence:

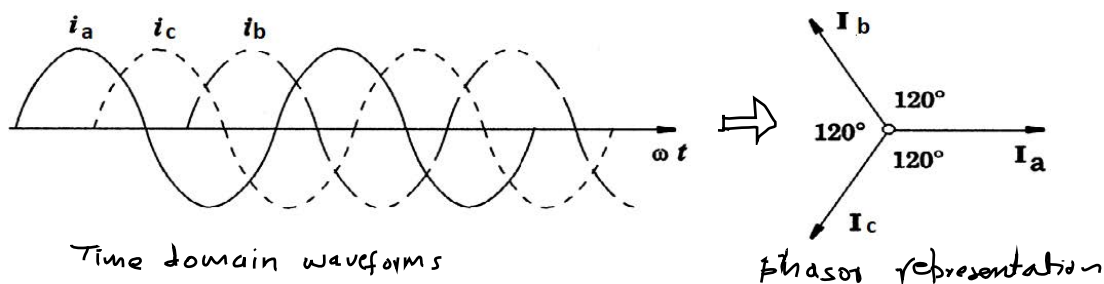
We have learnt the concept of phasors in year 1. This is a review to it with additional information on the 3-phase system.

Phase sequence: Order of occurrence of phasors:

- Either voltages or currents can be used. **Currents** in the following fig is in ABC sequence:



- Draw waveforms for **ACB** sequence. Show the phasors.



## Why Phase sequence is important to note?

When Paralling two voltage sources as it is done in utility (CEB) applications:

- The phase sequence should be the same.
- Other features required to represent a sinewave should be the same.

$$V(t) = X_m \sin(\omega t + \phi)$$

↑                      ↑                      ↑  
peak                      frequency                      phase

It is to be noted that we discussed systems in the steady state for this module hence the transient state is neglected in the analysis. However, it is important to note that the transient exist for a while and decay due to inherent resistive elements in the circuit, which come as an advantage for ac equipment operation.

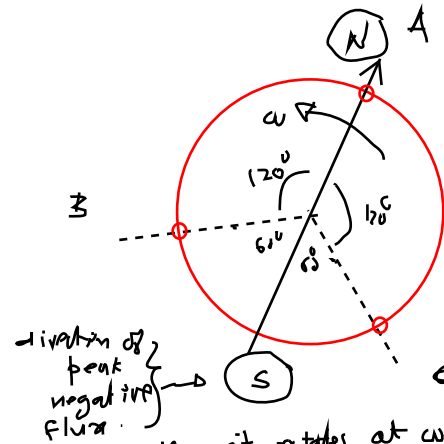
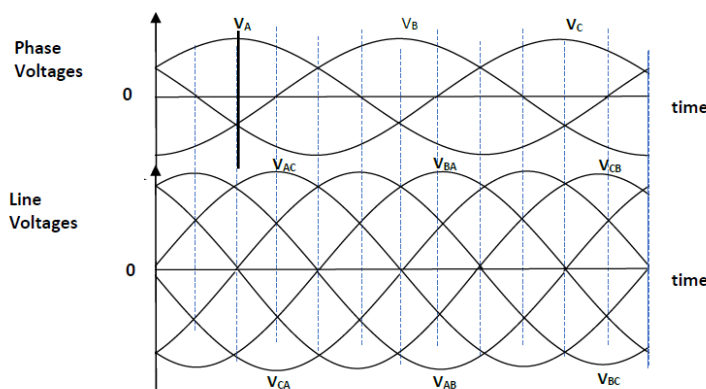
Excercise:

Explain why resistive elements assist healthy implementation of ac circuits. Use RL circuit with sinusoidal voltage source without a phase shift, and the switch in the circuit operates to connect the source to the load at  $t=0$ .



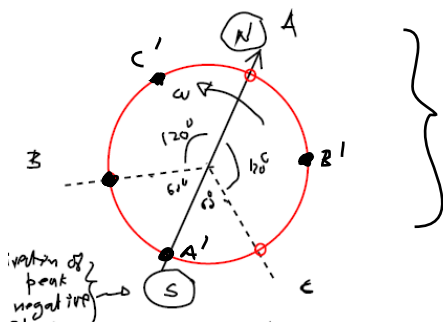
A balanced, 3-phase voltage set generates a 3-phase balance-current on a balanced load only. The connection of 3-phase coils can be done in two ways. The advantages of balance loading needs to be observed further.

- consider the figure below. A magnet with cosine flux distribution, rotates at  $\omega$  rad/sec



- The particular placement of conductors A, B, C generates a 3-phase balanced voltage set,

- now let us see how to form a phase A winding  $\rightarrow$  Phase A



conductor A and A' form  $180^\circ$  shifted wave forms

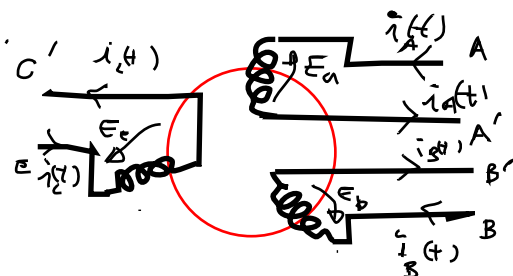


$$E_A = Z \phi_A \Rightarrow \text{phase A voltage}$$

- As an example, we can form 3-phase voltages using 3-concentric coils as above.

Note: In actual machines distributed coils are used in place of concentric coils due to various practical and technical reasons.

How to connect 3 coils for actual practice?:



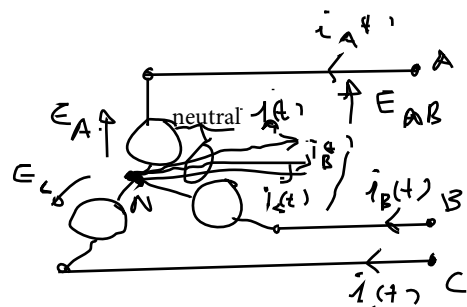
Star connected sources:

$E_{AN}$  - Phase Voltage (voltage at point A w.r.t neutral)  $\checkmark$  reads as

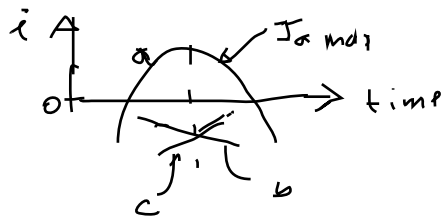
$E_{AB}$  - Line Voltage

$i_x(t)$  - Instantaneous current in phase x.

A', B', C' represent return conductors of each phase.

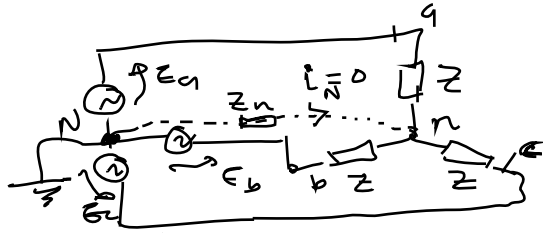


- For a balance load, the rms current of each phase is same. However, they are phase-shifted so that addition become zero.



$$i_N = i_a(t) + i_b(t) + i_c(t) = 0$$

- Therefore, for the balance system considered, the neutral current becomes zero. Therefore, Neutral conductor can be removed and a 3-phase load can be supplied with the identical electrical conditions as above.

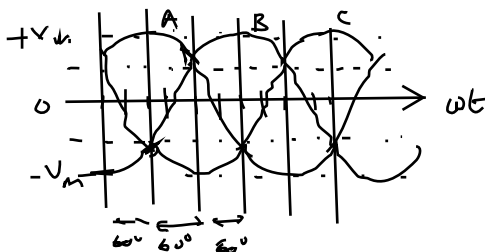


$$V_N = V_{nN} = 0 \text{ V}$$

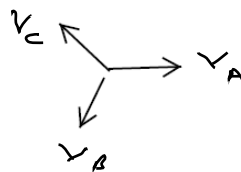
$Z_n$  - line impedance

Fig. 1  
Phase Sequence:  $\rightarrow$  ABC } Note  
 $\rightarrow$  ACB }

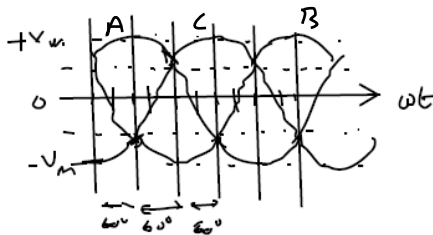
$\vec{V}_1$  leads  $\vec{V}$  by  $\alpha$   
 $\vec{V}_2$  lags  $\vec{V}$  by  $\theta$



ABC sequence (said to be +ve sequence)

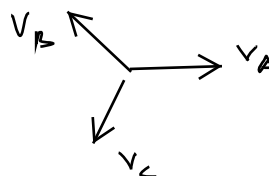


Phasor diagram



ACB sequence (said to be -ve sequence)

A leads C  
C leads B



## Review: Phasors

Sine cosine waveshapes with a fixed frequency can be represented by phasors in frequency domain.

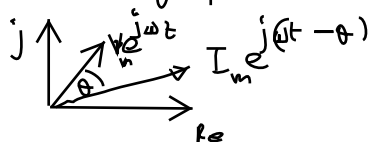


Fig. 2 Rotating Vectors (time domain)

at  $t=0$   
Frequency domain  
and

Phasor representation

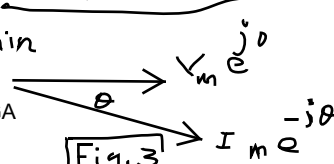


Fig. 3



## Notes:

- Projection of  $V_m e^{j\omega t}$  on Imaginary axis gives  $V_m \sin \omega t$
- Projection of  $I_m e^{j\omega t - \theta}$  on Imaginary axis gives  $I_m \sin(\omega t - \theta)$
- At  $t=0$ , the rotating vectors would occupy the position shown in Fig. 3 above.
- The rotating vectors above in Fig. 2 form two phasors as in Fig. 3.
- Phasors are represented by a capital letter with an upper bar, Ex:  $\bar{V}$ ,  $\bar{I}$
- A phasor has a magnitude and angle representing a phase angle.
- In power engineering, rms value is used to represent the magnitude.

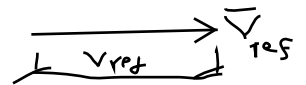
$$\bar{V} = V_{rms} \angle \phi \quad \Rightarrow \quad V_{rms} \sin(\omega t + \phi)$$

Frequency Domain Time Domain

In this case the reference phasor in time domain becomes,

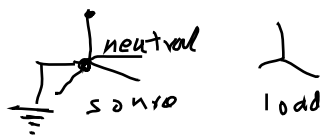
$$V(t) = V_{ref} \sin \omega t;$$

This voltage is represented by a horizontal line of length equal to  $V_{ref}$  in frequency domain as a phasor.

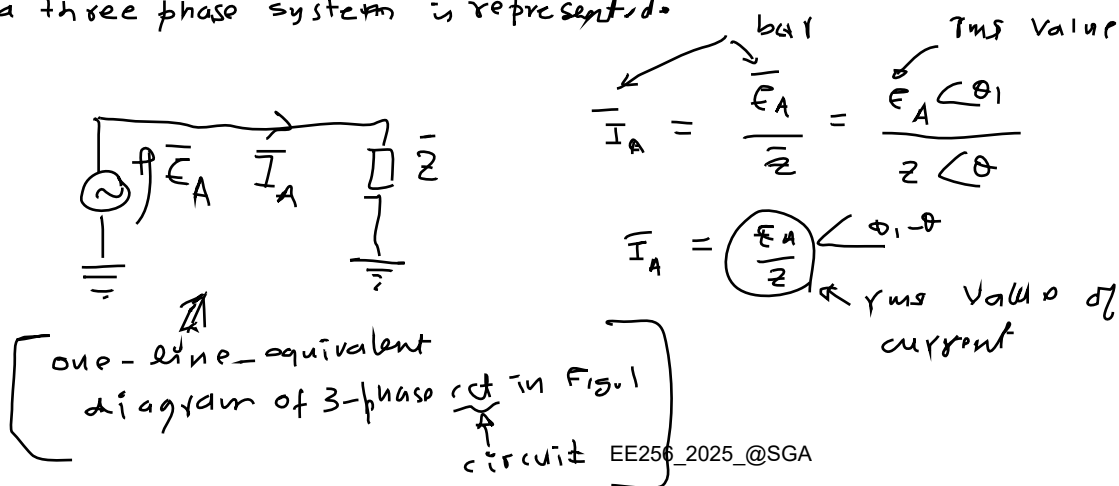


Analysis of 3-phase circuits:

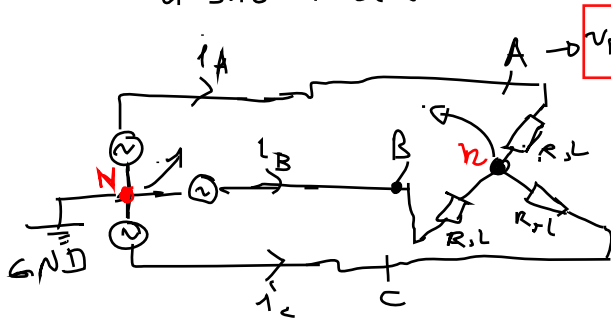
- Consider the 3 phase system connected in 'star' (Y) configuration.
- Usually the source is grounded. 3-phase loads typically have isolated neutrals.



If the load impedances ( $Z$ ) in each phase are equal, the load is said to be balanced. Then one-line diagram can be drawn for easy analysis of the circuit. Consider the Fig. 1 where, a three phase system is represented.



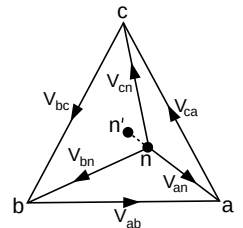
Quiz: 3-phase, 3 wire circuit with unbalanced load is shown below. Source is a balanced one.



$$v_n = R_n i_n + L \frac{di_n}{dt}$$

$Z_A \neq Z_B \neq Z_C$   
n - star point of the load

N' - star point of the source (typically grounded)



n'-balance load

\* comment about:  $V_{nN}$ ?

$$V_{nN} = \frac{V_A + V_B + V_C}{3} \text{ } \left. \begin{array}{l} \text{balanced} \\ \text{Load} \end{array} \right\}$$

$$V_{nN} = \frac{V_A + V_B + V_C}{3} \text{ for phases}$$

$$\frac{V_A(t) - V_{nN}(t)}{R_A} + \frac{V_B(t) - V_{nN}(t)}{R_B} + \frac{V_C(t) - V_{nN}(t)}{R_C} = 0$$

$$V_{nN}(t) \left( \frac{1}{R_A} + \frac{1}{R_B} + \frac{1}{R_C} \right) = \frac{V_A(t)}{R_A} + \frac{V_B(t)}{R_B} + \frac{V_C(t)}{R_C}; V_{nN} \neq 0 \rightarrow n'$$

Quiz:

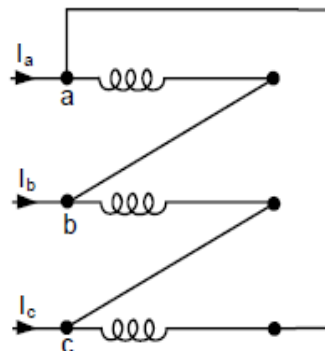
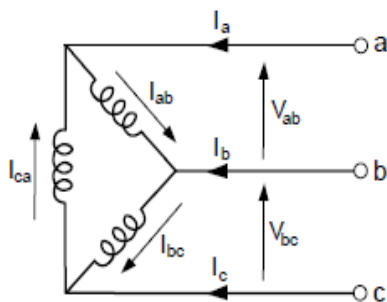
\* If N and n are connected using an ideal conductor ( $r = 0 \Omega$ ),  $V_{nN}$ ?

Will there be current in the connection,  $I_{nN}$ ?

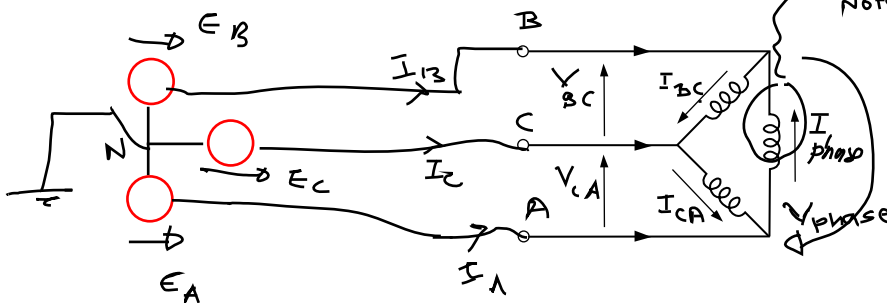
$$\text{Ans: } i_n = i_A + i_B + i_C \text{ (KCL) } \checkmark$$

Delta

\*  $\Delta$  connected loads:



One phase of the 3-phase load. Note the phase-voltage and phase current.



Phase currents:

- $I_{BC}$  flows from B to C
- $I_{CA} \rightarrow I_{AB}$

Quiz: Self study

Generators are typically connected in  $\Delta$  and grounded.

Find the reason for this:

Hint:

- Easy ground condition identification and protection.
- Lower insulation requirements.
- Avoid circulation currents and associated losses, due to unbalanced voltages.

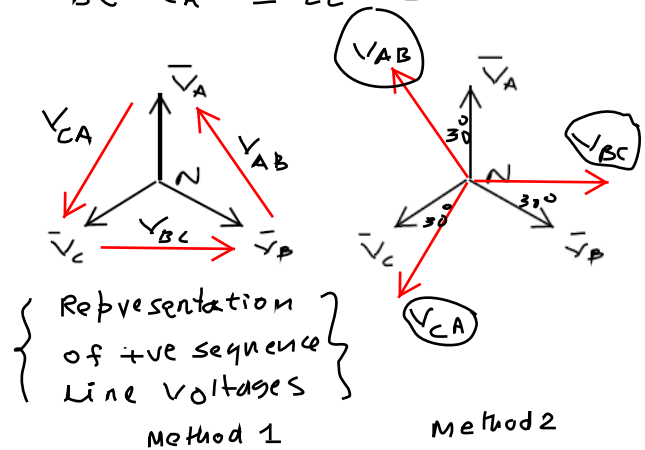
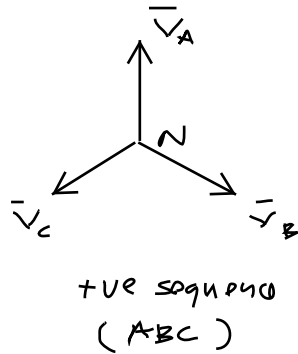
Q. Sketch the voltage phases. Phase to Neutral,  $V_{AN}, V_{BN}, V_{CN}$   
Line-to-Line  $V_{AB}, V_{CA}, V_{BC}$

△ connected load:

Load phase voltage = line-to-line voltage of the source  
Load phase current  $\neq$  line current of the source.

Consider ABC sequence for the source:

→ Then phase voltage set is  $V_{AN}, V_{BN}, V_{CN} \equiv V_{LN}$   
→ Then line voltage set is  $V_{AB}, V_{BC}, V_{CA} \equiv V_{LL}$  } source is connected



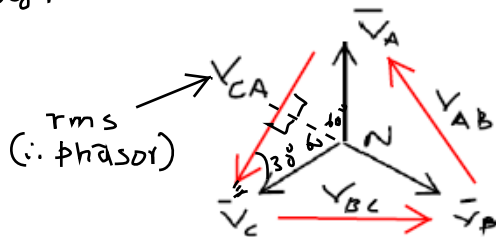
Observation:

\* Positive sequence line-voltage set leads positive sequence phase-voltage set by  $30^\circ$ .

\* Magnitudes:  $|V_{AB}| = |V_{BC}| = |V_{CA}| = |V_{LN}| \sqrt{3} = V_{LN} \sqrt{3}$

$V_{AB} = V_{AB}$  drop bar

Proof:



$$|V_{LL}| = |V_{LL}| = 2|V_{LN}| \sin 60^\circ$$

$$= 2 V_{LN} \frac{\sqrt{3}}{2}$$

$$V_{LL} = \sqrt{3} V_{LN}$$

∴ For ABC sequence (+ve sequence); positive sequence line voltage:

$$\vec{V}_{LL} = \sqrt{3} \vec{V}_{LN} \angle 30^\circ \text{ leading angle}$$

+ve sequence line voltage set      +ve sequence phase voltage set

Phase current of the load and phase voltage of the load.

Note: Balanced-source supplies balanced line current  $I_A, I_B, I_C$ .

How to sketch  $I_{\text{phase}}$ ?